1. Accuracy is an appropriate evaluation metric for imbalanced classification datasets.
   a) True
   b) False
   Correct Answer: b) False
   Explanation: Accuracy is not a suitable metric for imbalanced classification datasets because it does not properly reflect the performance of the model when one class dominates the dataset. Instead, metrics like Precision, Recall, F1-score, or AUC-ROC are more appropriate as they consider class imbalances.

2. Which metric is used to evaluate regression models by penalizing larger errors more than smaller ones?
   a) Mean Absolute Error (MAE)
   b) Mean Squared Error (MSE)
   c) Root Mean Squared Error (RMSE)
   d) R-squared (R^2)
   Correct Answer: b) Mean Squared Error (MSE)
   Explanation: MSE is a commonly used evaluation metric for regression that squares the differences between actual and predicted values. This squaring penalizes larger errors more heavily than smaller ones, making MSE more sensitive to outliers in the dataset.

3. What is the primary objective of simple linear regression?
   a) To classify data points into categories.
   b) To establish a linear relationship between a single independent variable and a dependent variable.
   c) To evaluate the clustering of data points.
   d) To measure the variance of the data.

   Correct Answer: b) To establish a linear relationship between a single independent variable and a dependent variable.
   Explanation: Simple linear regression aims to find a linear equation that best describes the relationship between an independent variable (predictor) and a dependent variable (target). The goal is to predict the dependent variable based on the independent variable using the equation $y = mx + c$, where m is the slope and c is the intercept.

4. Which type of regression is best suited for modelling non-linear relationships in data?
   a) Linear Regression
   b) Polynomial Regression
   c) Logistic Regression
   d) Ridge Regression
   Correct Answer: b) Polynomial Regression
   Explanation: Polynomial Regression extends linear regression by introducing polynomial terms to model non-linear relationships between the dependent and independent variables.

5. In polynomial regression, what happens when the degree of the polynomial is too high?
   a) The model may suffer from underfitting
   b) The model generalizes well to new data

c) The model may suffer from overfitting
d) The model complexity remains unchanged
Correct Answer: c) The model may suffer from overfitting
Explanation: A high-degree polynomial can lead to overfitting, meaning it captures noise in the training data rather than the underlying trend, reducing its ability to generalize to new data.

6. Which of the following feature is not provided by the Pandas module?
   a) Merge and join the dataset
   b) Filter data using conditions
   c) Plot and visualize the data
   d) None of the above
   Ans: c) Plot and visualize the data
   Explanation: Pandas module provides features like merge, join, filtering of datapoints in the dataset. Plot and visualization of the data can be done through matplotlib.pyplot package
7. Which of the following commands return the data type of the values in each column in the data frame df?
   a) print(df.type)
   b) print(df.datatype)
   c) print(df.dtypes)
   d) print(dtypes(df))
   Ans: c) print(df.type)
   Explanation: print(df.type) commands return the data type of the values in each column in the data frame
8. Which Pandas function is used to perform one-hot encoding?
   a) one_hot_encode()
   b) get_dummies()
   c) to_categorical()
   d) encode_categorical()
   Ans: b) get_dummies()
   Explanation: The get_dummies function converts categorical variables into dummy variables, which is essentially one-hot encoding, making it a convenient tool for preparing data for machine learning models that require numerical inputs.
9. What indicates that you have a perfect fit in linear regression?
   a) R2 > 0
   b) R2 = 0
   c) R2 > 1
   d) R2 = 1
   Ans: d) R2 = 1
   Explanation: In linear regression, an R^2 value of 1 indicates a perfect fit, meaning all data points fall exactly on the regression line and the model perfectly predicts the dependent variable based on the independent variable.
10. Which Python library provides the KFold and StratifiedKFold classes for cross-validation?
    a) TensorFlow
    b) Numpy
    c) Scikit-learn
    d) Pandas
    Ans: c) Scikit-learn

Explanation: Scikit-learn is a popular machine learning library in Python, and it includes various cross-validation techniques, with KFold and StratifiedKFold being readily available classes for implementing K-fold and stratified K-fold cross-validation respectively.